

from the OS in a standard fashion. RHEL 7 provides *mcelog* for logging hardware faults, which provides a framework for catching and logging machine check exceptions on x86 systems. On supported systems, it can also automatically mark bad areas of RAM so that they will not be used.

Install and enable *mcelog* as shown below:

```
# yum -y install mcelog.
# systemctl enable mcelog.
#systemctl start mcelog.
```

From now on, hardware errors caught by the *mcelog* daemon will show up in the system journal. Messages can be queried using the *journalctl -u mcelog* service. If the *abort* daemon is installed and active, it will also trigger on various *mcelog* messages. Alternatively, for administrators who do not wish to run a separate service, a cron is set up but commented out in */etc/cron.hourly/mcelog.cron* that will dump events into */var/log/mcelog*.

6. Memtest86+

When a physical memory error is suspected, an administrator might want to run an exhaustive memory test. In such cases, the Memtest86+ package must be installed. Since the memory test in a live system is more than ideal, the Memtest86+ package will install a separate boot entry that runs Memtest86+ instead of the regular Linux kernel. The following steps outline how to enable this in the boot entry:

- 1.) Install the Memtest86+ package and this will install the Memtest86+ application into */boot*.
- 2.) Run the command *memtest-setup*. This will add a new template into */etc/grub.d* to enable Memtest86+.
- 3.) Update the Grub2 boot loader configuration as shown below:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg.
```

7. Nmap

Nmap is an open source port scanner that is provided by the Red Hat Enterprise Linux 7 distribution. It is a tool that administrators use to rapidly scan large networks but it can

also do a more intensive port scan on individual hosts. Nmap uses the raw IP package in novel ways to determine what hosts are available on the network, what services those hosts are offering, what OS they are running, what type of packet filters/firewalls are in use and dozens of other characteristics.

The *nmap* package provides the *nmap* executable.

```
# yum -y install nmap
```

The example given in Figure 6 shows Nmap scanning the network. The *-n* option instructs it to display host information numerically, not using DNS. As Nmap discovers each host, it scans privileged TCP ports looking for services. It displays the MAC address, with the corresponding network adapter manufacturer of each host.

8. Wireshark

Wireshark is an open source, graphical application for capturing, filtering and inspecting network packets. It was formerly called Ethereal, but because of trademark issues, the project's name was changed. Wireshark can perform promiscuous packet sniffing when network interface controllers support it. RHEL 7 includes the *wireshark-gnome* package. This provides Wireshark functionality on a system installed with X.

```
#yum -y install wireshark-gnome.
```

Once Wireshark is installed, it can be launched by selecting *Applications > Internet > Wireshark Network Analyzer* from the GNOME desktop. It can also be launched directly from the shell using the following command:

```
# wireshark
```

Wireshark can capture network packets. It must be executed by the root user to capture packets, because direct access to the network interface requires root privileges. The capture option in the top level menu as shown in Figure 7 permits the user to start and prevent Wireshark from capturing

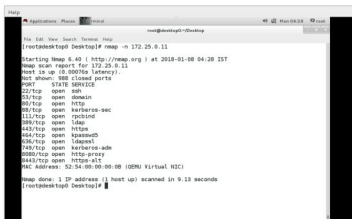


Figure 6: Nmap scanning ports of some other machine

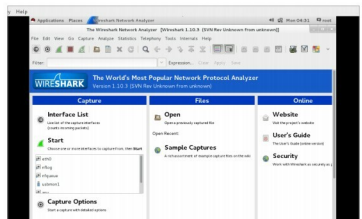


Figure 7: Graphical user interface of Wireshark